Keeping the customers worth keeping

A machine-learning churn early-warning proposal for Company A

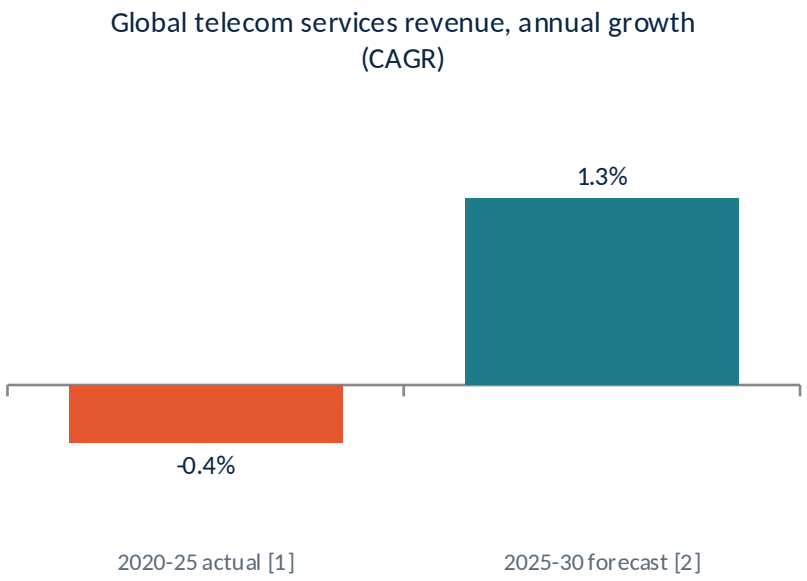
Proof-of-Concept proposal | GCI World 2026 Final Assignment | [Your name], November 2026

Executive summary: target the riskiest 20% of customers and retain an estimated $1.7M per 100,000 customers

Problem	Telecom revenue is nearly flat (-0.4% to +1.3% a year) [1][2], so every lost subscriber is expensive to replace. In Company A's sample, 49.6% of customers left within 31-60 days.
Evidence	Churn rises with handset age (42% to 58%), is highest in months 13-18, and is higher when usage falls sharply (55% vs 46%).
Model	Gradient boosting: ROC-AUC 0.695 on 20,000 unseen customers (logistic regression: 0.628). The top 20% riskiest customers are 73.9% churners vs 49.6% on average.
Proposal	Monthly risk scoring plus three targeted retention plays (handset refresh, usage win-back, month 13-18 check-in). Base case: +$1.7M net per 100,000 customers, +$0.5M over a simple "oldest handsets" rule. A 90-day A/B pilot comes first.

Sources: Company A dataset [6]; own analysis; market figures [1]-[5]. Impact figures rest on stated assumptions (slide 13).

Telecom revenue is barely growing, so growth has to come from keeping the subscribers you already have



$1,222B
Global telecom services market value in 2025 [1]

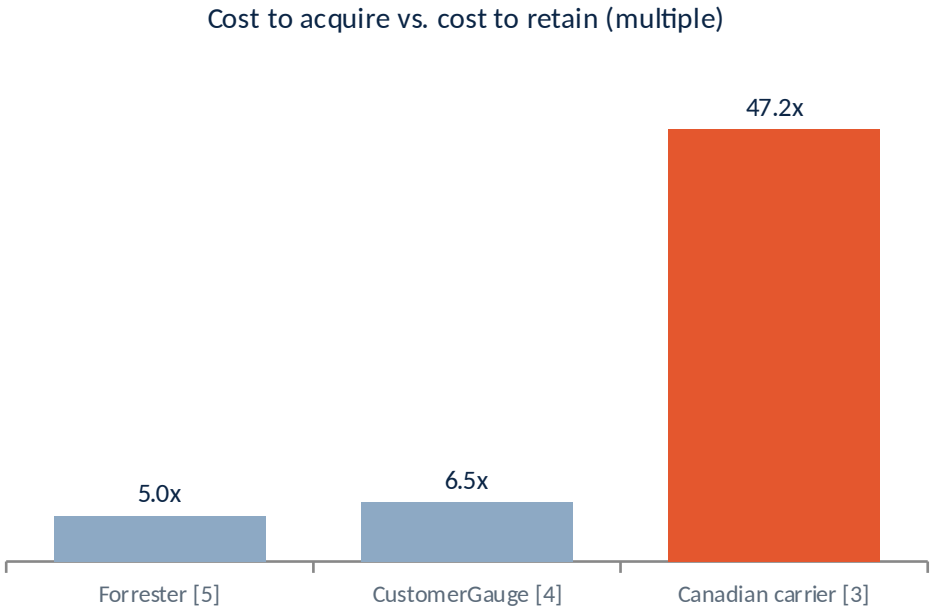
11.1B
Subscriptions in 2025, growing 3.6% a year in volume [1]: more lines, flat revenue = price pressure

5-50%
Typical annual churn: 20-50% overall [4]; 5-32% for postpaid [3]

Implication: with volume up and value flat, retention is the cheapest lever for protecting revenue.

Sources: [1] GlobalData/MarketLine via GII Research; [2] Analysys Mason (May 2026); [4] Tridens citing CustomerGauge 2024; [3] TM Forum via telecoms.com.

Acquiring a replacement customer costs roughly 5 to 50 times more than keeping one



What this means for Company A

- Acquiring a postpaid customer averaged C$521 in Canada [3]. A retention offer costing around $20 is small by comparison.
- Savings only appear if offers reach the right people. Contacting everyone wastes budget and trains loyal customers to wait for discounts.
- That is the job of a churn model: rank customers by risk so the offer budget goes where it changes outcomes.

Sources: [5] Forrester via AI Business; [4] CustomerGauge via Tridens (6-7x, midpoint shown); [3] TM Forum via telecoms.com (C$521 acquisition vs C$11.04 retention per customer, a 2017 Canadian carrier example).

Company A's data: 100,000 customers, 100 attributes and one clear outcome to predict

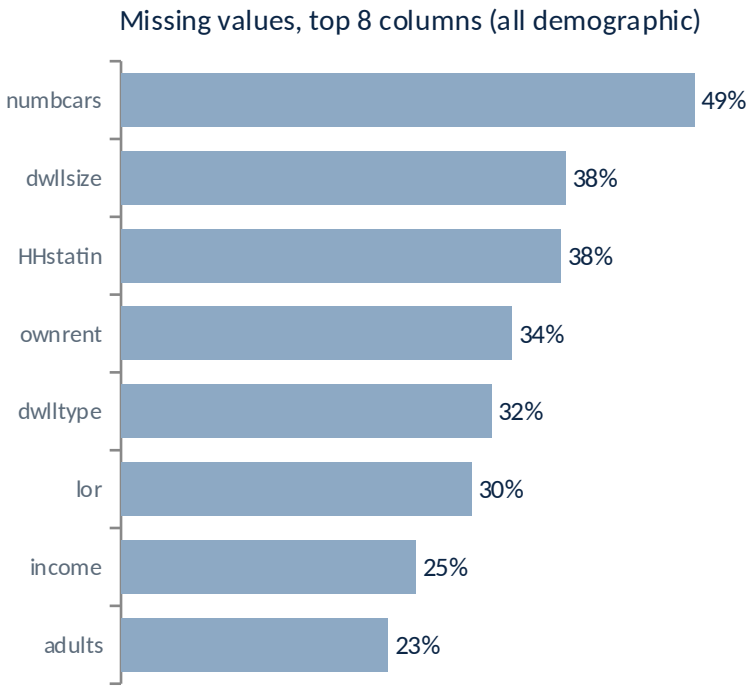
100,000
customers (one row each)

100
columns: usage, billing, handset, demographics

49.6%
churned 31-60 days after observation

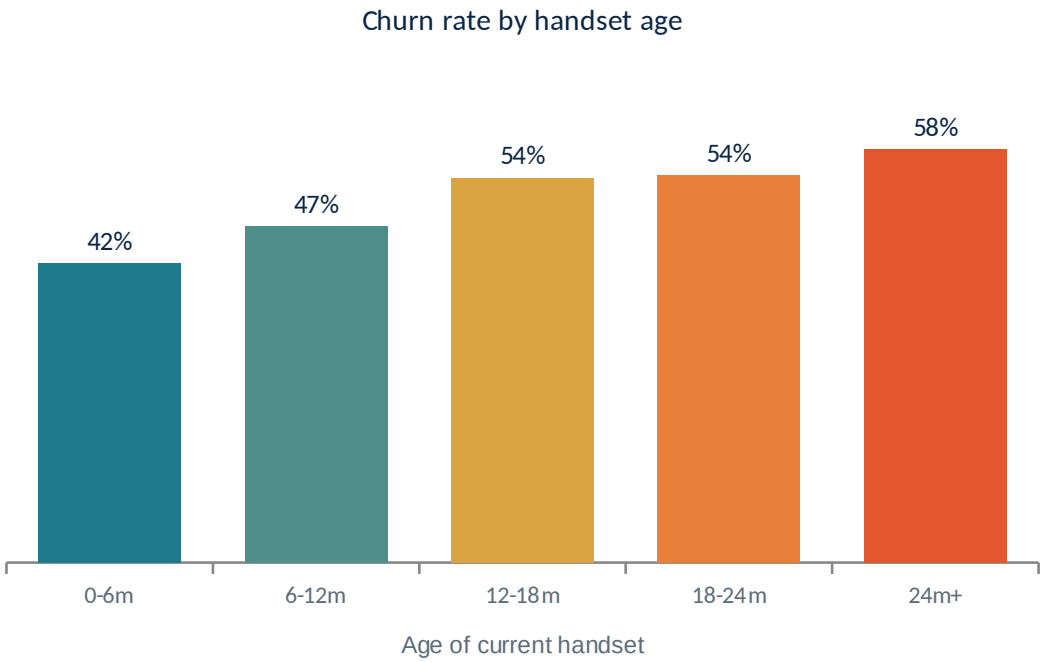
80 / 20
train / test split (stratified)

Heads-up: a 49.6% churn rate over one month is far above real-world levels, so the sample looks enriched with churners. We trust relative results (ranking, lift) and size the impact conservatively.



Source: Company A dataset [6] (Client.csv joined to Record.csv on Customer_ID; 1:1 match). Own analysis.

Customers with older handsets churn far more: 42% for phones under 6 months vs 58% beyond 24 months



363 vs 421

average handset age in days: stayers vs churners

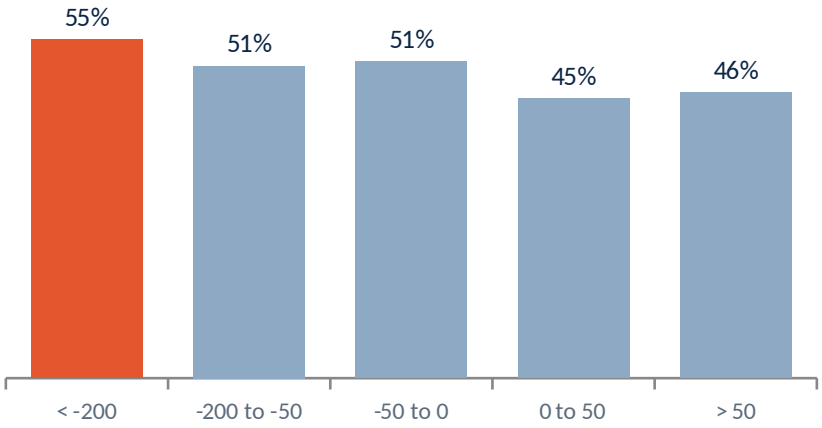
Early hypothesis

- Customers with aging phones are tempted by competitors' new-phone deals.
- Refurbished handsets also churn more (53.4% vs 48.9% for new).
- Action idea: offer timely upgrades before the customer shops around.

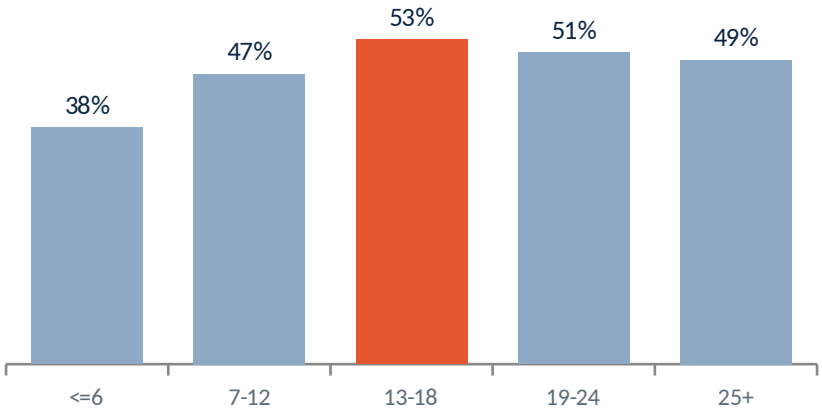
Source: Company A dataset [6], own analysis (eqpdays = age of current handset; n per bar between 11,296 and 36,356).

Churn peaks when usage is falling sharply and in months 13 to 18 of the customer relationship

Churn rate by change in usage (% vs prior 3 months)



Churn rate by months as a customer

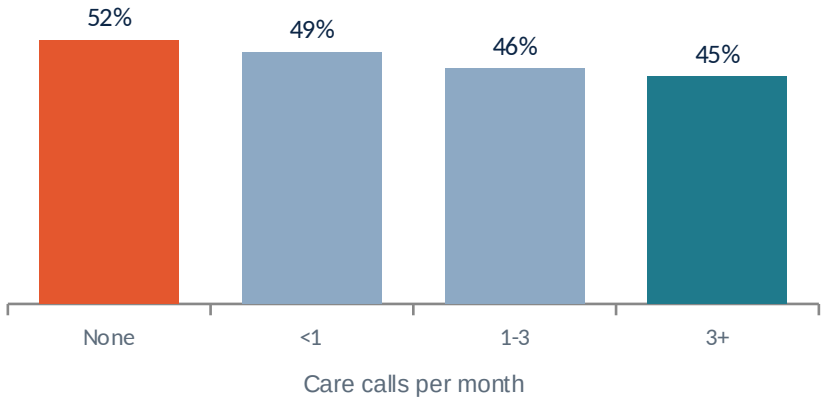


Falling usage is an early exit signal (55.0% churn vs 46.4% for growing usage). Churn peaks at 52.7% in months 13-18, around the end of a typical first contract.

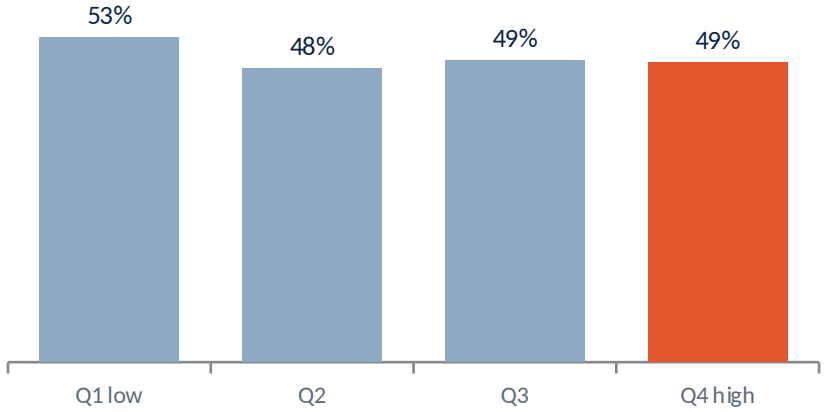
Source: Company A dataset [6], own analysis. change_mou = % change in minutes of use vs the previous 3-month average; months = months in service.

Silent customers churn most, and high-spending customers leave as often as low-spending ones

Churn by avg. monthly customer-care calls



Churn by monthly revenue quartile



Customers who never call care churn at 51.8% vs 44.7% for frequent callers: proactive outreach may help. The top revenue quartile churns at 48.6%, so valuable customers are at risk too.

Source: Company A dataset [6], own analysis. Associations only, not proof of cause. Regional gap: Northwest/Rocky Mountain 56.9% vs Midwest 45.9%.

From business goal to ML task: predict who will leave in the next 31-60 days, early enough to act

1. Business goal

Reduce revenue lost to customers leaving.

EDA shows churn is concentrated in identifiable groups, so targeting pays.

2. ML task

Binary classification: will this customer churn in the next 31-60 days?

Gives a ranked list the retention team can act on.

3. Target & inputs

Target: churn (1 = left, 0 = stayed). 110 inputs: 98 original + 12 engineered.

Engineered: usage trends, failed-call rate, handset age vs tenure, overage share.

4. Output

A 0-100% risk score per customer, refreshed monthly.

Contact the top 20% with the play that fits their profile.

Why churn rather than revenue or usage forecasting? It is labelled, tied directly to revenue, and actionable: a 30-60 day warning gives time to intervene.

Source: own design based on EDA and dataset description [6]. Features were built on training data only; the test set was held out.

Gradient boosting was the best model: ROC-AUC 0.695 on 20,000 customers it had never seen

Model	ROC-AUC	PR-AUC	Accuracy	Why we tried it
Logistic Regression	0.628	0.603	59.3%	Simple, explainable baseline
Random Forest	0.680	0.664	62.7%	Captures non-linear patterns; robust
Gradient Boosting (HistGB)	**0.695**	**0.684**	**63.8%**	Learns step-by-step from errors; best on tabular data

How to read the score

ROC-AUC is the chance that a customer who really churned gets a higher risk score than one who stayed. 0.5 = coin flip, 1.0 = perfect. We use it because the offer budget is limited, so ranking quality matters more than a yes/no cutoff.

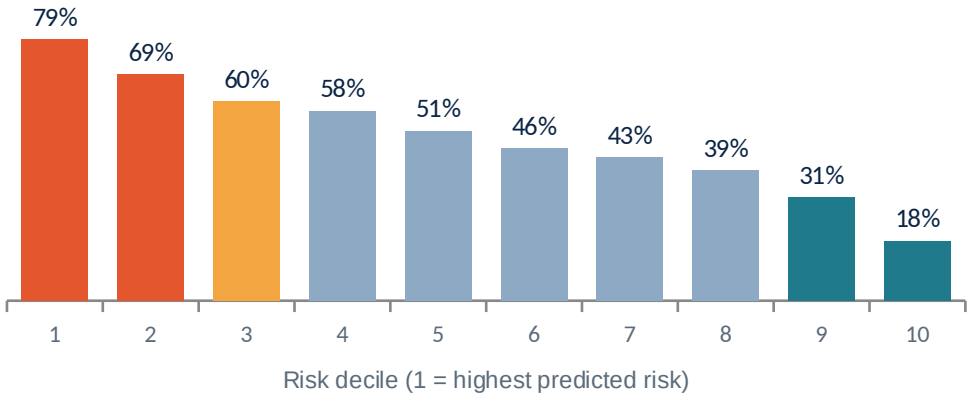
Honest take

0.695 is a moderate score: individual behaviour is hard to predict. The value is in ranking: the next slide shows the top-ranked customers churn far more than average. More behavioural history could lift performance further.

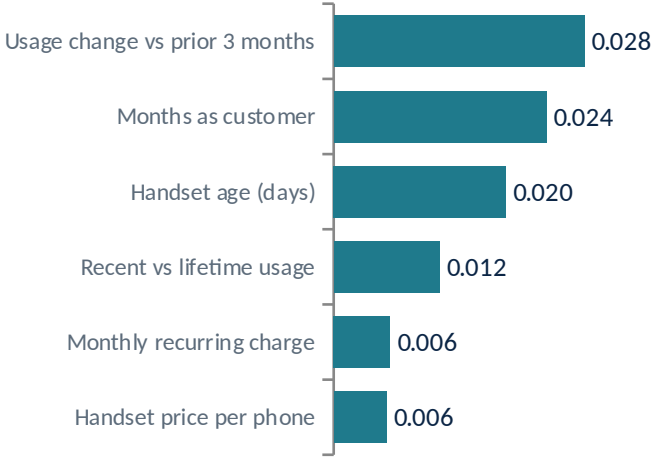
Source: own analysis; scikit-learn [7]. 80/20 stratified split (random_state=42). ROC-AUC is the primary metric; PR-AUC and accuracy for reference.

The model ranks customers well: the riskiest 10% churn at 79%, the safest 10% at 18%

Actual churn rate by predicted-risk decile



Top drivers of churn risk



73.9%
churners in the top 20% (vs 49.6% average)

29.8%
of all churners captured by contacting the top 20%

42.0%
of all churners captured by contacting the top 30%

Source: own analysis on the 20,000-customer test set; importance = drop in ROC-AUC when a feature is shuffled (permutation importance) [7].

Proposal: a monthly churn early-warning system that triggers three targeted retention plays

Handset refresh

Trigger: high risk + handset older than 12 months

Evidence: churn 54-58% vs 42% for new phones.

Offer: upgrade discount or trade-in credit, sent before the customer shops around.

Usage win-back

Trigger: sharp drop vs prior 3-month usage

Evidence: 55.0% churn when usage drops sharply.

Offer: plan right-sizing, bonus data or minutes, quality check on dropped calls.

Month 13-18 check-in

Trigger: high risk in months 13-18, or no care contact

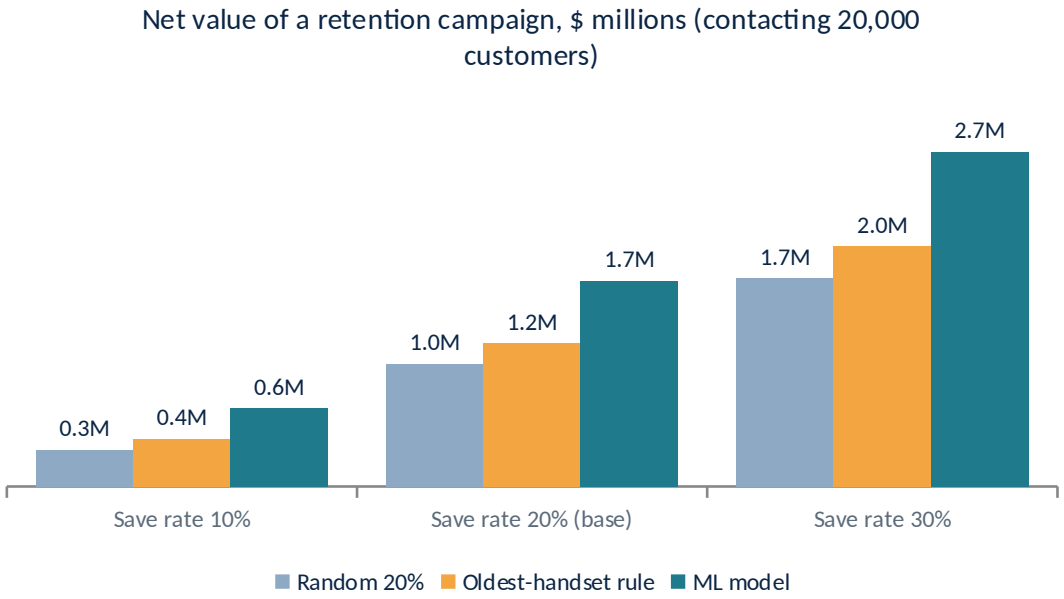
Evidence: 52.7% churn in months 13-18; silent customers churn 51.8%.

Offer: proactive care call and loyalty renewal offer.

Score all customers monthly → **Take the top 20% by risk** → **Assign the matching play** → **Compare with a control group**

Source: own proposal derived from EDA (slides 6-8) and model drivers (slide 11). Offers and triggers to be validated in the pilot.

Base case: +$1.7M net value per 100,000 customers, $0.5M more than a simple "oldest handsets" rule



Assumptions

- 100,000 customers; contact the top 20%
- Revenue per customer: $58.72/month (dataset mean)
- A saved customer stays 12 more months
- Offer cost: $20 per contacted customer
- Save rate: 20% base, 10-30% range (to be measured)
- Churn precision from the test set: 73.9% model, 55.8% rule, 49.6% random

Base case: model targeting nets $1.68M vs $1.00M random and $1.17M for the rule of thumb.

Source: own simulation on the test-set ranking (analysis.py). All economic inputs are assumptions to be validated in the pilot; value = retained revenue, not profit.

Next step: a 90-day A/B pilot proves the save rate before any large-scale rollout

Weeks 1-4

Prepare

Build the monthly scoring pipeline; recalibrate the model on Company A's true churn rate; define offers and a treatment/control split.

Weeks 5-12

Pilot

Contact half of the top-risk 20% with the matching play; the other half is the control group. Track churn and offer cost.

Month 4+

Scale

If net value is positive, roll out to all customers; retrain monthly; add new plays and richer data.

Key risks and how we handle them

- Enriched sample: real churn is lower, so recalibrate and re-measure lift on the full base.
- Correlation is not cause: the control group measures what offers truly change.
- Discounting loyal customers: cap offers; use the risk score and customer value.
- Drift and privacy: retrain monthly; limit use of demographic fields (up to 49% missing).

Success metrics

- Churn: treated vs control
- Net revenue retained
- Offer cost per saved customer

Source: own implementation plan. Timeline is indicative.

References

[1] GlobalData / MarketLine. Telecommunication Services Global Industry Almanac 2021-2030 (market value and CAGR, subscriptions). GII Research. https://www.giiresearch.com/report/mkl2137308-telecommunication-services-global-industry-almanac.html

[2] Analysys Mason (12 May 2026). Global telecoms market: trends and forecasts 2025-2030. https://analysysmason.com/research/content/regional-forecasts-/global-telecoms-forecast

[3] telecoms.com. Churn is breaking the telecoms market: here's how to fix it (summarising a TM Forum Quick Insight Report). https://www.telecoms.com/oss-bss-cx/churn-is-breaking-the-telecoms-market-here-s-how-to-fix-it

[4] Tridens Technology. Why Telecom Customers Churn and How to Measure it? (citing CustomerGauge, 2024). https://tridenstechnology.com/telecom-churn/

[5] AI Business / DataRobot webinar. Using AI to predict customer churn in the telecommunications industry (citing Forrester). https://aibusiness.com/verticals/using-ai-to-predict-customer-churn-in-the-telecommunications-industry

[6] Company A telecom dataset (Client.csv, Record.csv) and Dataset Overview, provided by GCI World 2026 course materials.

[7] Pedregosa et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research 12, 2825-2830.

[8] Generative AI use: Claude (Anthropic) assisted with analysis code and slide drafting. Chat history URL: [paste your Claude chat link here].